

A wind tunnel study on the effect of thermal stability on flow and dispersion of rooftop stack emissions in the near wake of a building

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HIGHLIGHTS

- ▶ Study on flow and dispersion of rooftop stack emissions in the near wake region.
- ▶ Varying atmospheric stratifications were stable, neutral and unstable conditions.
- ▶ Vertical velocity increases under stable conditions.
- ▶ Longitudinal turbulence intensity increases under unstable conditions.
- ▶ Concentration increases with stable condition and decreases with unstable condition.

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ABSTRACT

The effect of atmospheric thermal stability on wind flow and dispersion of rooftop stack emissions in the near wake of a building in an urban environment is investigated using stratified wind tunnel experiments. Experiments with varying stratifications (stable: Ri_b and $Re_b = 0.023$ & 5.1×10^3 , neutral: Ri_b and $Re_b = 0.0$ & 5.8×10^3 , and unstable: Ri_b and $Re_b = -0.016$ & 6.6×10^3) were conducted using an isolated cubical building model. Tracer gas (C_2H_4) was released from a stack on the building roof. Flow, temperature and dispersion fields were analyzed and measured using a split film, a cold-wire thermometer, and high-response flame ionization detector. Thermal diffusion was examined at different distances downwind of the leeward side of the building model. Flow and dispersion characteristics studied included mean velocity, turbulence intensity, mean and fluctuation concentrations in the near wake of the building. The results indicate that when atmospheric stratification is unstable, flow separation decreases and longitudinal turbulence intensity increases in the near wake region. When this stratification is stable, vertical velocity and vertical turbulence intensity increase in the near wake region. There is an increase in concentrations with stable conditions, and a decrease in concentrations with unstable conditions in the near wake region.

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1. Introduction

A large number of industrial, institutional, university and hospital laboratories, as well as manufacturing facilities, emit a wide range of potentially harmful pollutants (e.g., toxic and odorous chemicals) from rooftop stacks in the urban environment. Therefore, the dispersion of potential hazardous exhaust from these stacks is of great concern when addressing possible consequences of such releases on human health and safety, as well as the

environment in the stack vicinity. Many factors affect this dispersion, such as wind velocity and behavior, topographic features around the site, stack height, buildings near the stack, and thermal stability (Wilson, 1979; White and Stein, 1990; Schulman and Scire, 1991).

Regarding the last factor, it is known that thermal stability is most important in the transport and dispersion of rooftop stack pollutants. The near-ground atmosphere typically exhibits a stable stratification during nighttime and unstable stratification during daytime. For moderate to higher wind speeds, however, sufficient atmospheric turbulence is generated such that near-neutral stability prevails.

A number of relevant wind tunnel experiments have been performed to investigate various influences, such as stack height and

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relative distance between stack and intake on pollutant dispersion, as it proceeds from the stack to intakes. Li and Meroney (1983) carried out wind tunnel experiments on a building model to assess effluent concentration in the near wake region. The results showed that effluent concentrations were reduced by the presence of a sharp edge on the building. The majority of previous research on rooftop stack emissions has been done for buildings (e.g., White and Stein, 1990; Schulman and Scire, 1991; Thompson, 1993; Wilson et al., 1998; Petersen et al., 2002; ASHRAE, 2007, 2009, 2011; Chavez et al., 2011). These studies have shown that an increase in stack height or exhaust speed decreases roof-level concentrations. Furthermore, Lowery and Jacko (1996) studied the effects of building parapets and varying intake configurations on the dilution of a pollutant between a rooftop stack and building intake. The parapets were found to produce a cavity zone that extends above the roof by as much as twice the physical height of the parapet; increasing stack height had little effect on dispersion until the stack extended beyond this cavity region.

Recently, Saathoff et al. (2009) evaluated the effect of rooftop structures (RTSs) on the dispersion of building exhaust, using wind tunnel experiments with an isolated building for four different full-scale equivalent heights, from 15 to 70 m. The influence of the RTS on plume behavior for normal wind diminished as building height increased, because of an increase in size of the separated flow region above the roof. Hajra et al. (2011) examined the effects of near-field pollutant dispersion of upstream buildings in the built environment and compared them to the ASHRAE (2007) model. The effect of spacing (S) between the buildings and stack from the upwind edge of the emitting building (X) were also investigated. Data show that within the recirculation zone, a change in along-wind dimension of the upstream building has a negligible effect on the dilution of emissions from the downwind building. Plume geometry is largely governed by the upwind dimensions of the upstream building. On the other hand, the effect of near-field pollutant dispersion in the case of downstream buildings in the urban environment was studied in a wind tunnel by Hajra and Stathopoulos (2012). The results showed that the height and a crosswind dimension of the downstream building, as well as spacing between buildings, are critical parameters for assessing plume dilution. The effect of wind direction, exhaust speed, stack location, stack height, and RTS crosswind width on severity of the downwash effect of an RTS, representing a typical RTS effect on plume dispersion, was assessed by Gupta et al. (2012). Downwash produced by the RTS significantly increased roof-level concentration, depending on building height, stack location, stack height, exhaust speed, wind direction and RTS crosswind width.

Besides wind tunnel experiments, numerical modeling is an important tool to investigate the rooftop stack emissions. Zhang et al. (1996) examined the potential application of the $k-\epsilon$ model TEMPEST for simulating flow field and dispersion patterns around a cubical building under stably stratified conditions, while Flowe and Kumar (2000) investigated the feasibility of using $k-\epsilon$ model FLUENT as a means of modeling air flow past a building and stack geometry to develop parameterizations useful to air quality modeling. Ahmadi and Li (2000) developed a computational model for Lagrangian simulation of particle transport, dispersion and deposition near a small isolated building. Smith et al. (2001) used a three dimensional (3D) computational fluid dynamics (CFD) model, coupled with a meteorological radiation and surface physics package, to model the mean flow field and tracer dispersion in the vicinity of an idealized cubical building. Olesen et al. (2009) recently used a comprehensive dataset on dispersion behind rectangular buildings to analyze the performance of two dispersion models in respect to their handling of building effects: the Danish OML model and the US AERMOD model with the PRIME building

algorithm; additionally, the German MISKAM model has been assessed. The degree of misprediction in respect to the maximum concentrations is less than a factor of two for OML and AERMOD. Lateb et al. (2011) investigated the dispersion of pollutants exhausted from a building roof stack located in the wake of a tower is by means of the realizable $k-\epsilon$ turbulence model. The result shows that increasing the stack height has an effect similar to that obtained by increasing the momentum ratio, but with some differences, depending upon which wall of the two buildings is considered.

The effect of thermal stability on pollutant dispersion in an urban environment has been the subject of numerous investigations. However, there are limited experimental studies of the effect of atmospheric stability on dispersion within the wake of a building. Yang and Meroney (1970), Kothari et al. (1985), Higson et al. (1995) and Mavroidis et al. (1999) studied dispersion in the building wake under different atmospheric stratifications. Uehara et al. (2000) examined in a wind tunnel the effects of thermal stratification on flow in and above urban street canyons. Yassin et al. (2005) experimentally investigated pollutant dispersion over a built-up area in the field, and in a wind tunnel under different thermal stabilities and wind directions. The results indicate that thermal stability and wind direction are important parameters in the study of atmospheric diffusion. Santos et al. (2009a,b) examined in the field the effects of averaging time on building-influenced atmospheric dispersion, under different atmospheric stabilities. Very few experimental studies have dealt with the effect of atmospheric stability on dispersion.

Also, very few of these studies suggest a strong effect of thermal stability around an isolated buildings, and there is no work treating the effect of thermal stratification on dispersion patterns of rooftop stack emissions. Consequently, there is a need for further study of the effect of thermal stability on pollutant dispersion. Given the knowledge gained from prior research, the goal of this study is to conduct a wind tunnel investigation of wind flow and pollutant dispersion emitted from rooftop stacks in the near wake of a building. We also examine how this dispersion is affected by various types of thermal stability, using a stratified boundary layer wind tunnel.

2. Experimental approach

2.1. Thermally stratified wind tunnel

An open circuit, thermally stratified boundary layer wind tunnel at Tokyo Polytechnic University, Japan, was used under the following atmospheric stabilities: stable, neutral and unstable. The wind tunnel working section is 1.2×1.0 m and length is 16.0 m (Fig. 1). The wind tunnel consists of a fan, temperature profile cart

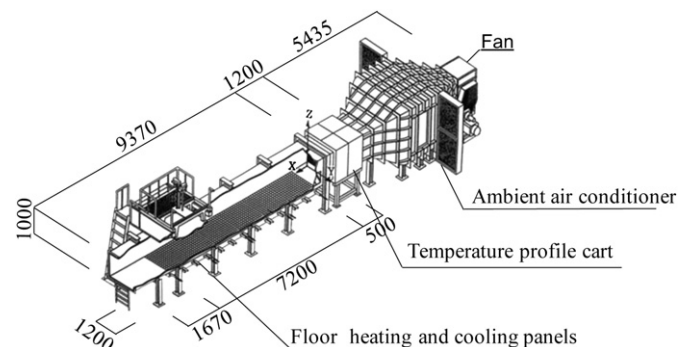


Fig. 1. Thermal stratified wind tunnel.

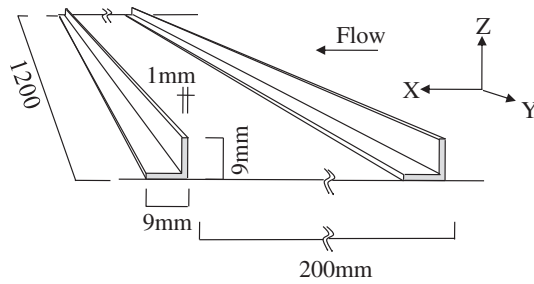


Fig. 2. Aluminum plates as roughness elements.

to measure vertical temperature, floor heating and cooling panels, and ambient air conditioner. Air temperature is controllable in the range 10–50 °C. The experiments were conducted with a wind velocity and air temperature at inlet of 1.3 m s⁻¹ and 15 °C.

2.2. Urban boundary layer simulation

A simulated urban boundary layer was achieved using 26 thin aluminum plates, 9 mm high, and placed at 200-mm pitch on the wind tunnel floor (Fig. 2). This combination of plates produced a simulated atmospheric boundary layer with a normal depth $\delta = 500$ mm and a free stream wind speed $U_\infty = 1.3$ m s⁻¹. A widely used model to describe the vertical velocity distribution in the urban environment, in which the boundary layer is fully developed, can be described by a power law:

$$U_Z/U_H = (Z/H)^n, \quad (1)$$

where U_Z is mean velocity at elevation Z , U_H is mean velocity at the building height, and n is the power exponent. We used the typical value $n = 1/4$ for an urban area. However, there is a range of other possible values for n . For details on typical values for different terrain types, refer to Snyder (1981). Fig. 2 shows the simulated turbulent boundary layer in the stratified wind tunnel under the three atmospheric stabilities (stable, neutral and unstable) at $X = 0.0$. Fig. 3 shows typical temperature profiles in the vertical

Table 1

Bulk Richardson number (R_b), building and stack Reynolds numbers (Re_b & Re_s) and flow/dispersion characteristics in stratified thermal stability.

Atmosphere stability	R_b	Re_b	Re_s	Flow characteristics	Dispersion characteristics
Neutral	0.0	5.8×10^3	1.2×10^2	Vertical velocity lower	Increase concentration close to the ground
Stable	0.023	5.1×10^3	1.03×10^2	Vertical velocity increases	Increase concentration
Unstable	-0.016	6.6×10^3	1.3×10^2	TKE increases	Decrease concentration

direction for a stable and unstable stratified boundary layer at $X = 0.0$. In stable and unstable conditions, the profiles seem to be similar slightly changed from neutral condition. This is because of the R_b is about 0.016 for stable conditions and about -0.23 for unstable conditions (Table 1), meaning weakly stable and weakly unstable, respectively. Wind profiles show small variation up to 500 mm in Fig. 4, except for the region near the floor, where flow is highly turbulent because of the effect of roughness. The stratified flows show almost linear temperature profiles in the streamwise direction.

2.3. Urban building model

The three-dimensional isolated cubic building model was 100 × 100 × 100 mm. The model was constructed of steel. The simulated scale of the model in the urban boundary layer was 1:300. The model was located at $X/H = 0.0$. The stack was modeled by a steel cylinder of height 7 mm, with inner and outer diameters 4 and 6 mm. The stack was set at the center of the building roof. Fig. 5 shows a schematic of the building with stack.

2.4. Flow and temperature measurements

A split film was adopted for wind velocity measurement, and a 5-μm cold-wire thermometer (Fig. 6) used for temperature measurement. To precisely compensate for the contribution of

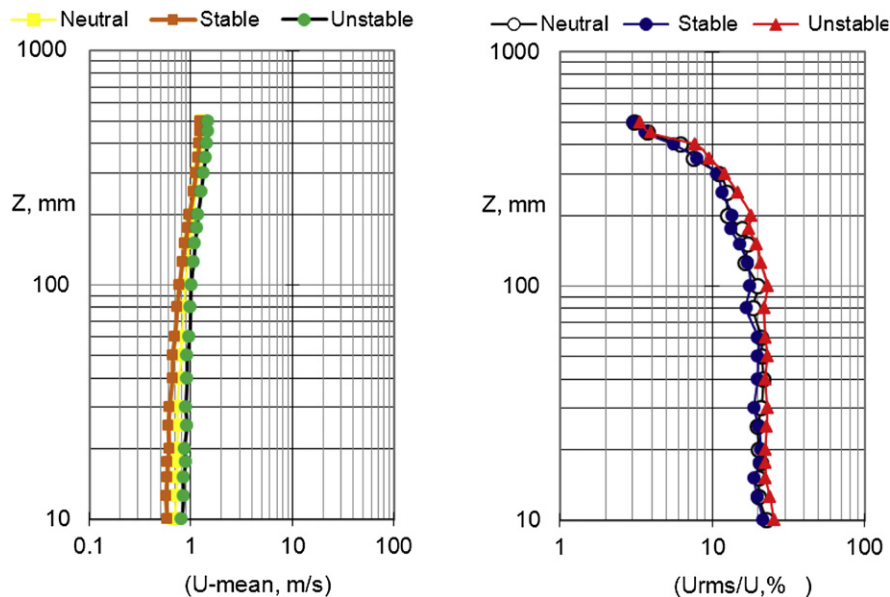


Fig. 3. Vertical distributions of mean velocity and turbulence intensity in the simulated boundary layer under thermal stability.

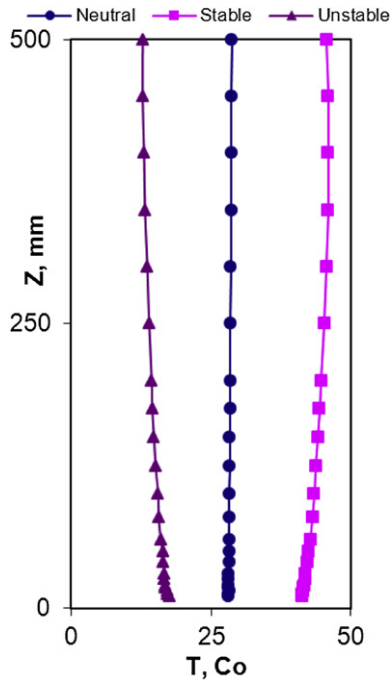


Fig. 4. Vertical profile of temperature in the simulated boundary layer.

temperature fluctuation to output voltages of the split film under the low wind velocity condition, it was necessary to obtain precise calibration data under the stable wind velocity and temperature condition, thereby improving calibration precision and efficiency. The split film and cold-wire were set adjacent at distance $\Delta Y = 2$ mm, respectively, to measure fluctuating velocity and temperature simultaneously. The sampling frequency and low-pass filter cutoff frequency were set at 1000 Hz and 200 Hz, respectively, and 60,000 data points were sampled in 60 s.

2.5. Pollutant concentration measurement

The stack gas used as a tracer gas in diffusion measurement was ethylene (C_2H_4), without buoyancy. Thus, buoyancy effects were negligible. A high-response hydrocarbon analyzer detector (flame ionization detector, FID) was used to measure C_2H_4 concentration. Each measurement was made for 90 s and produced an average value. The background was also recorded. The difference between the local and background reading was taken as the final concentration measurement. However, the value of background concentration is nothing respectively, if compared to the mean concentration value. Concentration measurements are presented as the general non-dimensional concentration $K = C/C_0$, where C is measured concentration and C_0 the reference concentration

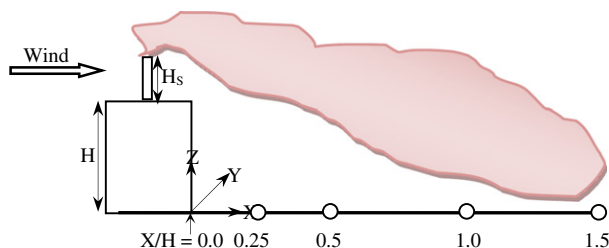


Fig. 5. Building and stack models.

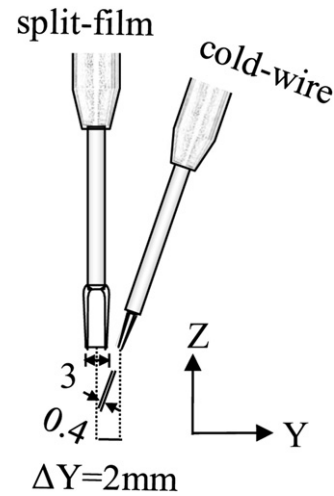


Fig. 6. Layout of sensors.

($C_0 = Q/U_H H^2$), in which Q is the source volume flow rate and U_H free stream velocity at building height H . Exhaust momentum from the stack was 50% of the free stream velocity. Since the density of C_2H_4 gas is nearly that of air, the density of pollutant gas can be thought to have this same density at the height of the pollutant effluent in the boundary layer.

3. Simulation criteria

Simulation criteria for wind tunnel experiments on atmospheric diffusion primarily include the following: (a) Geometry of the topography; (b) the ratio of pollutant release speed to that of the air flow, and atmospheric thermal stability; and (c) surface boundary layer characteristics. Further, the three parameters representing the ratio of inertial properties to molecular properties are: (1) Reynolds number Re ; (2) Péclet number; (3) Schmidt number. For more details on the similarity criteria, see Snyder (1981, 1985), Saathoff et al. (2009) and Gupta et al. (2012).

3.1. Atmospheric thermal stability criteria

The three different thermal stratifications (stable, neutral and unstable) within the atmospheric boundary layer were created in the test section by controlling the wind flow temperature and wind tunnel floor temperature. The bulk Richardson number Rb was selected to accurately represent atmospheric thermal stability from a practical point of view. This is because for Richardson number Ri in wind tunnel experiments, there was wide variation in wind velocity and temperature values below the building height ($Z < H$), so stability was indicated instead by Rb

$$Rb = \frac{gH(T_H - T_S)}{T_S U_H^2}, \quad (2)$$

where H is the reference height, T_H is temperature at H , and T_S is temperatures at the ground level. The resulting Rb in thermal stability is shown in Table 1. The value of Rb in the current experiment was sufficient to create the thermally stratified boundary layer.

3.2. Flow and dispersion criteria

Wind tunnel modeling criteria for accurate flow and dispersion in the near wake of a building were studied in detail by Snyder

(1981). The following criteria are generally suggested for gaseous exhausts without buoyancy under neutral stratified conditions in wind tunnel dispersion studies Gupta et al. (2012):

- Geometric similarity
- Similarity of wind flow in wind tunnel to the atmospheric surface layer
- Building Reynolds number ($Re_b = U_H H / \nu$) should be larger than 1.1×10^4
- Stack Reynolds number ($Re_s = U_s d_s / \nu$) should be larger than 2×10^3
- Equivalent stack momentum ratio ($M = U_s / U_H$),

where ν is the kinematic viscosity of air, U_H is the wind velocity at building height, d_s is outlet diameter of the stack, and U_s is exit velocity of the stack. Furthermore, Castro and Robins (1977) and Snyder (1994) suggested Re_b should be larger than 4×10^3 . Resulting Reynolds numbers based on Re_b and Re_s in thermal stability are shown in Table 1. On the other hand, Hout and Weil (1972), Lape (1987) and White and Stein (1990) showed Re_s as low as 300, 700 and 533 were adequate for a buoyant plume, respectively. In wind tunnel experiments for pollutant dispersion it is very important to maintain turbulent flow around the building and stack. Resulting Reynolds numbers based on Re_b and Re_s in thermal stability are shown in Table 1. Despite the low value of Re_s , a trip was not used in the stack. The low exhaust momentum caused the plume to be trapped in a separate flow region and consequently, the effect of laminar exhaust flow should be negligible. Furthermore, Saathoff et al. (1995) suggested that “it is generally not possible to satisfy the stack Reynolds number for small diameter stacks and it is also difficult to trip the flow for such stacks.” Although the stack Reynolds number was somewhat lower than 2000, this may have had minimal effect on measurement results, as discussed in Hajra et al. (2011).

4. Result and discussion

To evaluate the effects of thermal stability on flow and pollutant dispersion characteristics in the near wake of the building, we did wind tunnel experiments using the cubic building model with stack on the roof. Measurements were made at four different downwind locations along the wind tunnel centerline, $X/H = 0.25, 0.5, 1.0$ and 1.5 (Fig. 5). Flow and pollutant measurements in the turbulent boundary layer were made at various heights, beginning at 10 mm above the ground of the wind tunnel.

4.1. Mean velocity components measurements

Mean velocity components were normalized by the obtained mean velocity U_H , which is the approaching wind velocity at building height H in the thermal atmospheric boundary layer. Fig. 7 shows contours and vectors of mean velocity in the longitudinal component in the near wake of the building, under the stable, neutral and unstable atmospheric conditions. The vertical and lateral profiles of mean velocity in the longitudinal and vertical components under different atmospheric stratifications are shown in Figs. 8–10.

From the figures given, the formation of recirculation zones in the near wake region cannot be observed, but the pattern shows that the velocities in the lee building side are decreasing. Therefore, the flow movement in the near wake region appeared dispersion. Evans in 1957 (Hosker, 1984) introduced an equation to calculate the reattachment length (X_r) for buildings. The geometry of the buildings was characterized by length (l), height (H) and width (W). The equation can be applied for buildings with $l/H < 2$.

$$\frac{X_r}{H} = \frac{l}{H} + \frac{A(W/H)}{1.0 + B(W/H)} \quad (3)$$

Where; $A = -2.0 + 3.73.7.(l/H)^{-1/3} B = -0.15 + 0.305.(l/H)^{-1/3}$

The reattachment length (X_r) equal $0.19H$. It means that a small recirculation zone can be found in the near wake of the building and flow movement was weak. Therefore, the length of leeward recirculation zone was slightly less than $0.2H$ respectively. In the physical modeling experiments, Thompson and Lombardi (1977) reported that flow separated at the leading edge of the top of the building model, resulting in flow reversal just above the top of the cube, and that the flow did not reattached to the cubical building roof.

In particular, when thermal stability was strong, wind velocity in the lower part of the near wake of the building was very low, dropping to 1 or 2% of the wind tunnel velocity. Longitudinal velocity increased rapidly upward, starting at $0.6 > Z/H < 1$ and gradually reaching $Z/H > 1$. The velocity increase at roof level was from flow shear. The change of longitudinal velocity was much larger in the near wake region at $X/H = 0.25, 0.5$, and 1 . This is from the substantial effect of turbulent mixing, which is created by flow distortion in the wake region. This makes a small separation in this region. A thin vertical profile was generated at $X/H = 1.15$. Flow patterns appeared symmetric in both edges of the lee side of building at $Z/H = 0.5$. This result is consistent with those found in previous studies by Snyder (2005) and Gupta et al. (2012).

The vertical velocity clearly indicates an upward motion for the flow. The change of vertical velocity was larger at $X/H = 0.5, 1$ and 1.5 . This is from the strong effect of turbulent mixing, created by flow distortion in the downwind direction. The vertical velocity was increased and directed upward at $X/H = 0.5, 1$, and 1.5 , because of the steep gradient of vertical turbulence intensity. The change of vertical velocity at $X/H = 0.25$ was small. This is because of less intense vertical turbulence.

The features mean velocity distributions under the three different thermal stabilities are as follows:

(a) Stable atmospheric stability:

Since the bulk Richardson number Rb was small, the atmospheric stratifications were slightly stable. The Re_b and Re_s were smaller than under unstable and neutral conditions. The stable profiles show decreased longitudinal velocity at this lower height near the ground surface. This is because of the stable condition is slightly. The lateral distributions increased of longitudinal velocity at height $Z/H = 0.5$. In general, mean velocity in the vertical direction for stable conditions was higher than that in unstable conditions (Snyder, 1981; Panofsky and Dutton, 1984; Tennekes and Lumley, 1994; Zhang et al., 1996), because: (a) heat flux is positive because of upward heat transfer; (b) production of turbulent kinetic energy is increased; (c) buoyant production is positive; (d) bulk Richardson number is > 0.0 . In addition to turbulence being smaller in stable conditions than in unstable conditions, the most important influence observed was in the vertical direction (Ogawa et al., 1981; Zegadi et al., 1994). Furthermore, stable condition should increase the downward motion in the velocity field, which tends to reduce the recirculation region. The reattachment length of the separated flows at $X/H = 0.25$ was longer than that at the other locations. Vertical velocity increased because of there is less vertical momentum in the wake region resulting in a smaller separation region.

(b) Unstable atmospheric stability:

The Re_b and Re_s were higher than under stable and neutral conditions. The unstable conditions increased the longitudinal

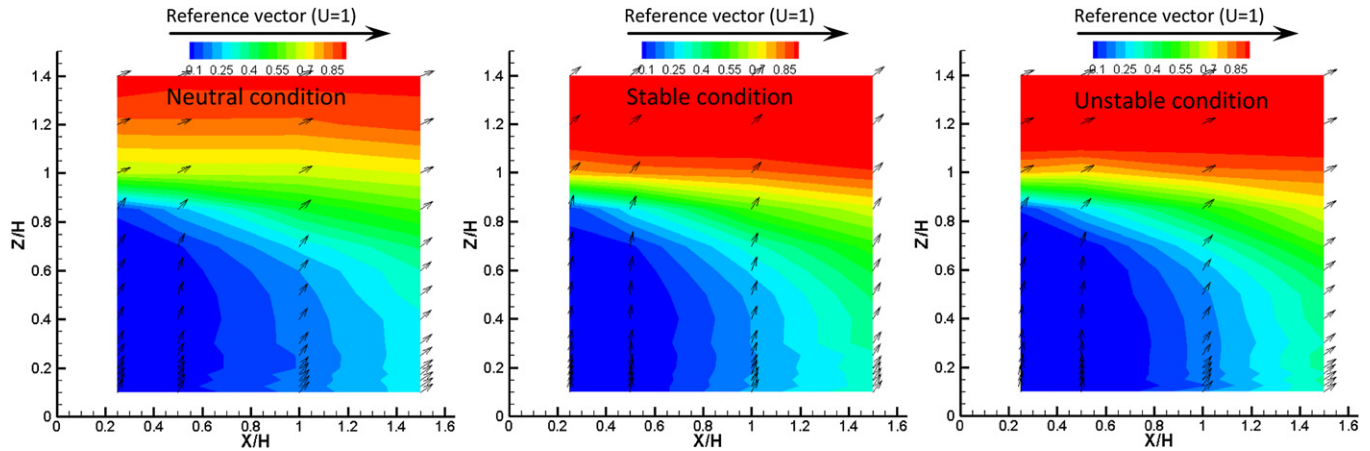


Fig. 7. Contours and vectors of mean velocity in the near wake of a building.

velocity at lower height $Z/H = 0.6$. This is caused by increased turbulence, which augments momentum transfer from higher to lower levels. The flow separation in the wake region is smaller than under stable and neutral conditions. The length of the recirculation region in the wake region is longer. The lateral profiles of longitudinal velocity decreased at $Z/H = 0.5$. Vertical velocity was lower than under stable condition.

(c) Neutral atmospheric stability:

The Re_b and Re_s were higher than under stable condition. The flow momentum was transformed into lateral and vertical momentum. The neutral conditions decreased the longitudinal velocity at height $Z/H = 0.6$. The lateral distributions of longitudinal velocity show the minima at the centerline in the near wake region. The neutral profiles show decreased longitudinal velocity at this lower height than under unstable condition. The flow separation in the wake region is higher than under stable condition. Vertical velocity was lower than under stable and unstable conditions.

The results show that when atmospheric stratification is stable, vertical velocity increases in the near wake of the building. While, longitudinal velocity decreases at the lower height near the ground surface. This is because of the atmospheric stratification was

slightly stable. When this stratification is unstable, vertical velocity and flow separation decrease in the near wake region. On the other hand, when stratification is neutral, longitudinal velocity is larger than under stable conditions in the near wake of the building.

4.2. Turbulence intensity measurement

The effects of atmospheric stratification on the longitudinal and vertical turbulence intensities are very significant and strongly affect the velocity field and concentration distributions. Vertical profiles of normalized turbulence intensity in the longitudinal and vertical components are shown in Figs. 11 and 12. Peaks of longitudinal and vertical turbulence intensities were observed in its upper portion, near the stack. This turbulence increased upwards gradually, and reached a maximum near the roof surface. This is because of shear production by the turbulent intensity. While, minimum longitudinal and vertical turbulence intensities were found in the lower portion of the near wake.

Variations of longitudinal turbulence intensity in atmospheric stratifications were strong far from the stack at $X/H = 1.0$ and 1.5 . The longitudinal turbulence intensity changed direction, with upward motion at $X/H = 0.25, 0.5$ and 1.5 starting at $Z/H > 0.8$ in the wake region, and a constant downward motion in the cavity owing

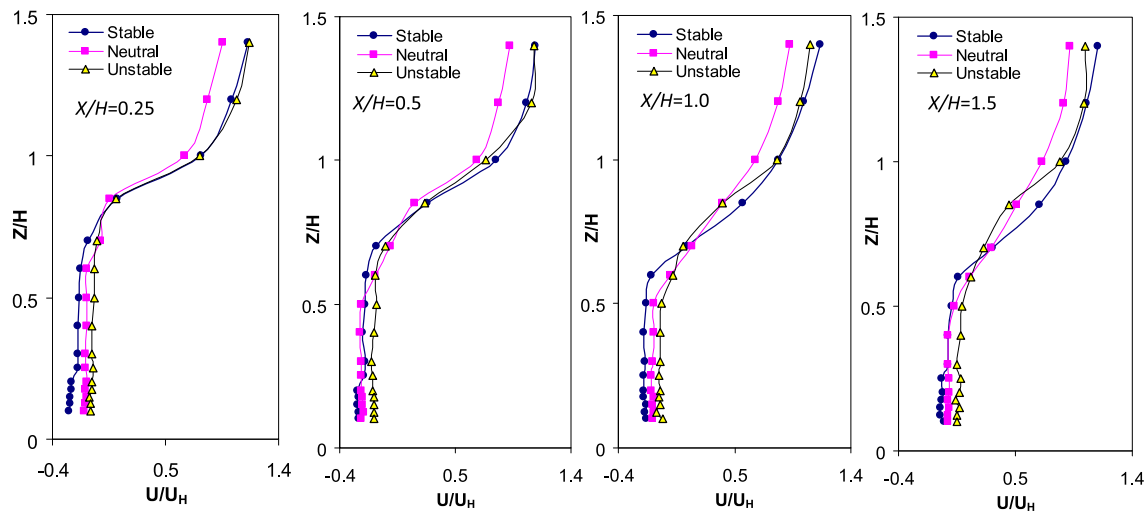


Fig. 8. Vertical profiles of mean velocity component in the longitudinal direction, U/U_H

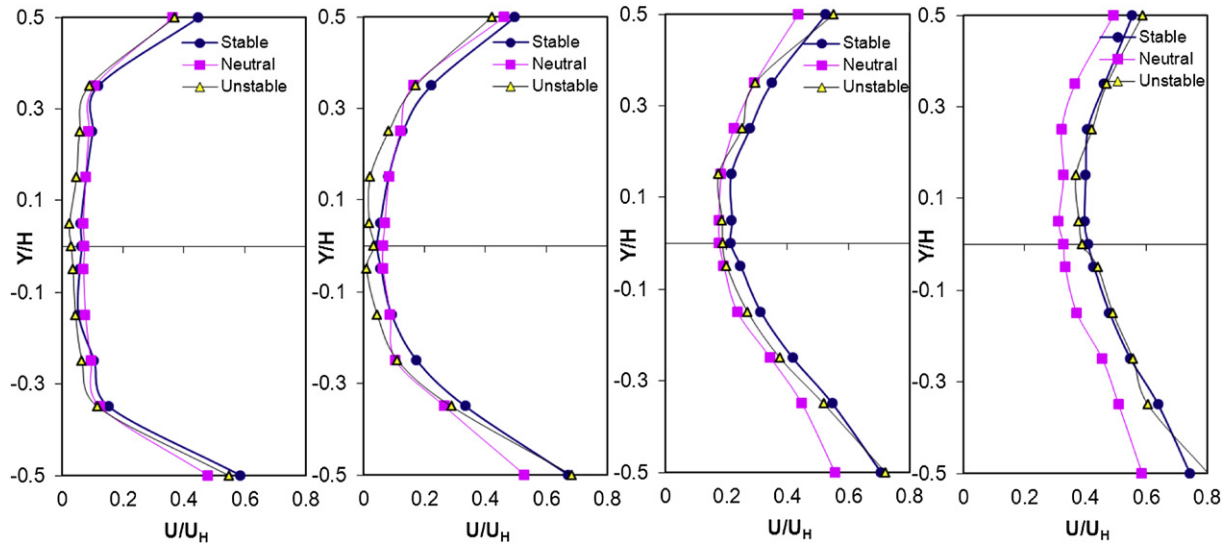


Fig. 9. Lateral profiles of mean velocity component in the longitudinal direction, U/U_H at $Z/H = 0.5$.

to a lack of momentum exchange and a weak vortex. Longitudinal turbulence intensity was much lower in the turbulent mixing layer at $X/H = 0.25$ and 0.5 . The effect of thermal stability on longitudinal turbulence intensity was much larger with increasing distance from the building. This result shows that as longitudinal turbulence increases, a smaller portion of the plume is carried upwind because of a reduction in the size of the reverse flow region.

Greater vertical turbulence intensity always generates a larger turbulent spread of the plume, and the size of the region of flow influenced by the building also affects plume spread. It is seen that the high vertical turbulence is even stronger close to the roof surface. This behavior is mainly related to intense temperature gradients in this region, which create strong buoyancy effects. Thickness changes of the vertical turbulence intensity in the boundary layer at $X/H = 0.5, 1$ and 1.5 were approximately the same, whereas they increased at $X/H = 0.25$. The variations of vertical turbulence intensity with thermal stability were nearly as much as near the ground surface. The effect of thermal stability on this turbulence was much larger with distance from the building model.

The features turbulence intensity distributions under the three different thermal stabilities are as follows:

(d) Stable atmospheric stability:

Longitudinal turbulence intensity under stable condition was higher than under neutral conditions and lower than under unstable conditions. The vertical turbulence intensity in stable conditions was higher than that in neutral and unstable conditions.

(e) Unstable atmospheric stability:

Longitudinal turbulence intensity in unstable conditions was observed to be higher than those in stable and neutral conditions. The longitudinal turbulence intensity in unstable conditions was much higher at $X/H = 1$ and 1.5 , because of a large velocity gradient. Vertical turbulence intensity was much higher at $X/H = 1$ and 1.5 , because of a large velocity gradient. On the top of the building, the peak values of vertical turbulence intensity was observed.

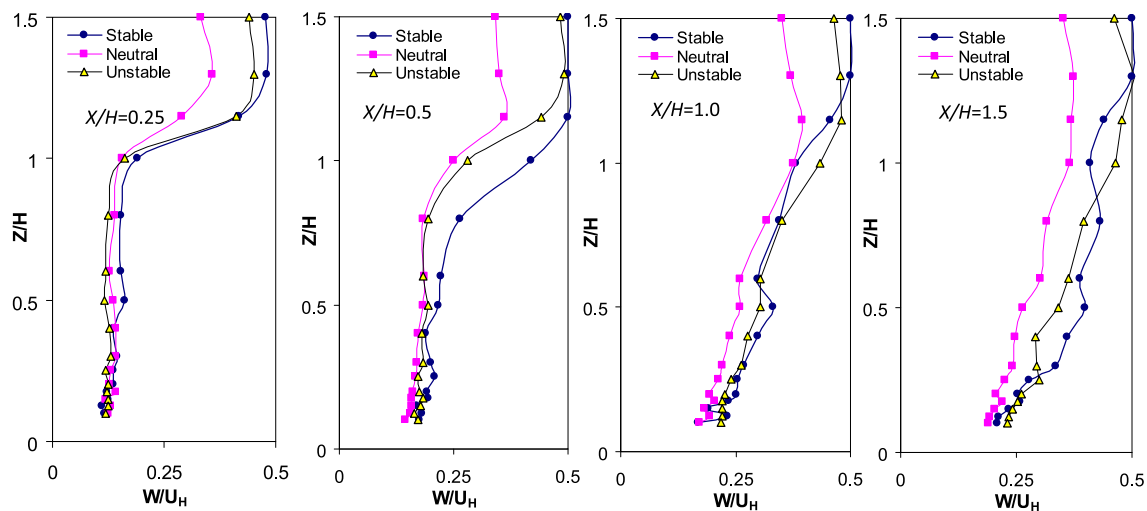
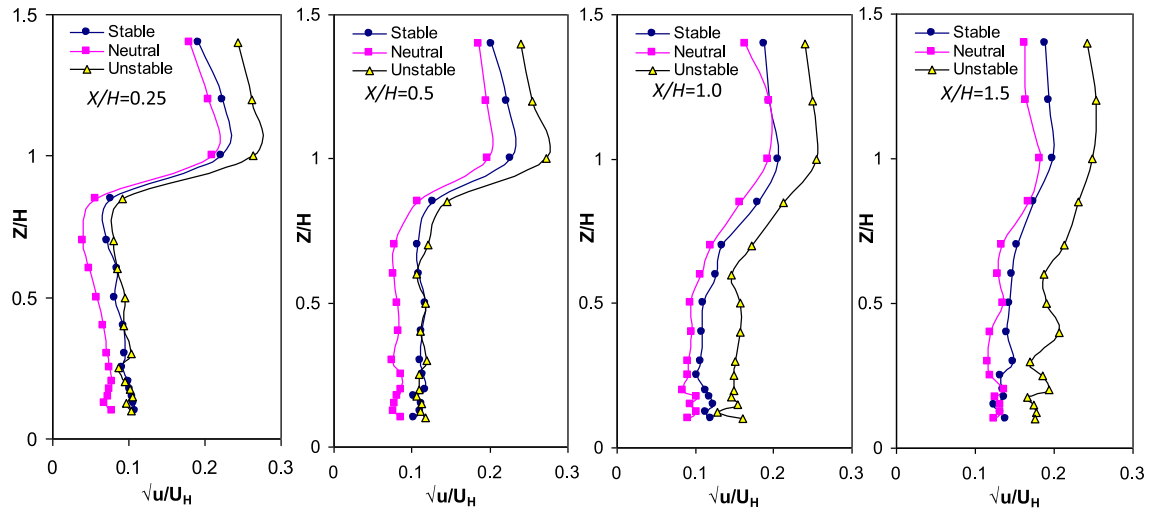


Fig. 10. Mean velocity component in the vertical direction, W/U_H

Fig. 11. Longitudinal turbulent intensity, $\sqrt{u}/U_H$

(f) Neutral atmospheric stability:

The longitudinal and vertical turbulence intensities observed to be lower than those in stable and unstable conditions in the near wake of the building. On the top of the building, the values of vertical turbulent intensity was higher than under stable condition.

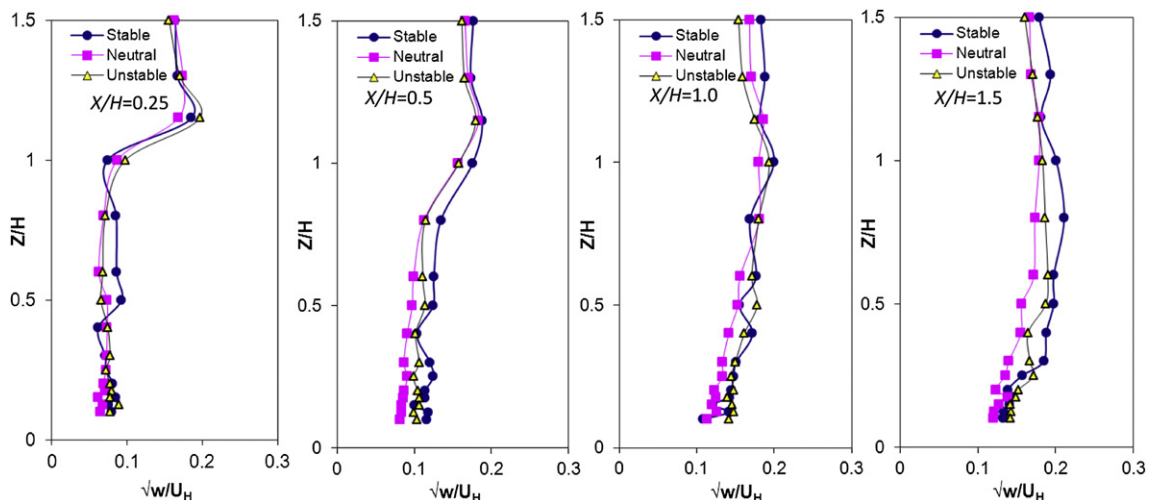
4.3. Mean and fluctuation concentration measurements

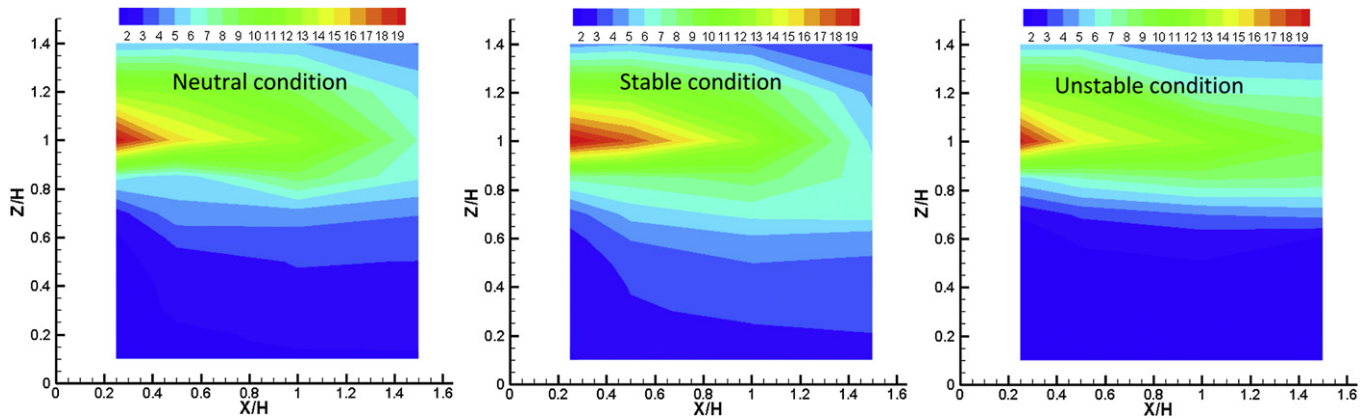
To establish dispersion characteristics of the simulated boundary layer, concentration measurements were made through the thermal stability in the near wake of the building. The stack model was located at the center of the building roof. Fig. 13 shows contour lines concentration measured in the near wake of the building, under the stable, neutral and unstable atmospheric conditions Figs. 14 and 15 shows vertical and lateral distributions of concentration measured in the thermal boundary layer.

For different thermal stratifications, dispersion patterns varied significantly downstream of the buildings. Near the buildings, however, dispersion was dominated by mechanical turbulence

generated by the building. In particular, when effluents emerge from the vertical stack at low momentum or low mean vertical velocity, and horizontal flow around the stack is sufficiently strong, the effluent plume may be drawn down in the low pressure region in the near wake of the stack. This phenomenon is referred to as stack downwash (Ogawa et al., 1974; Arya, 1999). The vertical distributions of concentration clearly show downwash in the near wake of the building. They are highly skewed in the upward direction, and the negative tail is essentially in the downward direction for $Z/H < 1$. In spite of the highly non-Gaussian character of the concentration distributions close to the stack, they tend toward Gaussian a short distance downwind in the near wake of the building. Moreover, it is clear that the centroid of these distributions descends slightly with downwind distance. This result appears consistent with the observations of Snyder and Lawson (1991).

Concentration distributions gradually increased upward starting at $0.75 < Z/H < 1$, reaching a maximum at $Z/H = 1.0$, and gradually decreased at $Z/H > 1.0$. Peak concentrations of thermal stability were observed at $Z/H = 1$, where the effluent was emitted near the

Fig. 12. Vertical turbulent intensity, $\sqrt{w}/U_H$

Fig. 13. Contour lines of mean concentration, K .

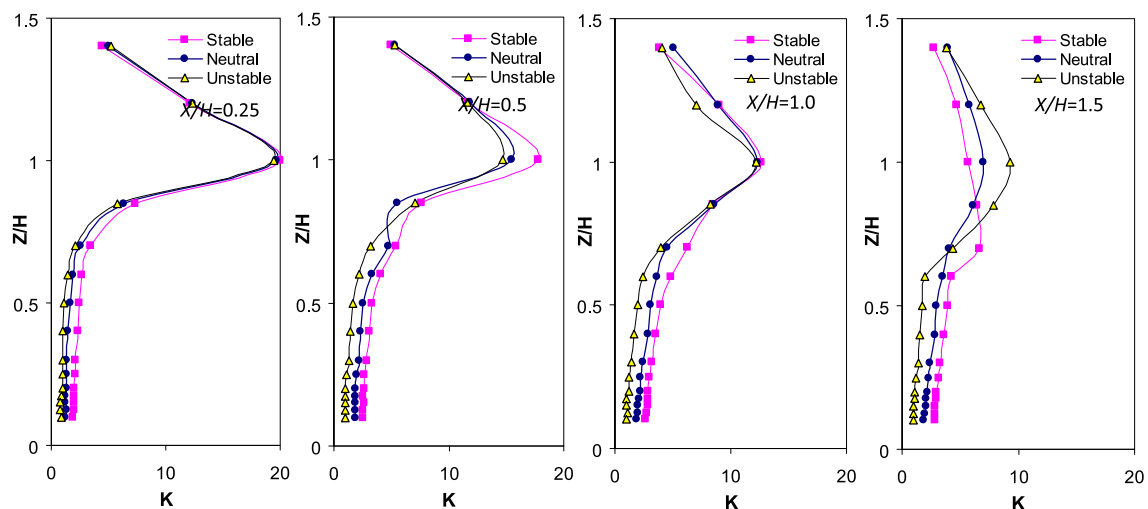
separation–reattachment region and created the downwash. This is because emission momentum from the stack was less than the free stream velocity. The maximum concentration was at $X/H = 0.25$, near the stack emission on the roof surface. The minimum concentration on the roof surface was at $X/H = 1.5$, far from the stack emission. Therefore, the dispersion concentration was high near the stack emission and decreased with distance from the stack. On the other hand, minimum concentration in the near wake was at $X/H = 0.25$. This is from the higher vertical turbulent intensity. Pollutant concentration in the near wake region at $Z/H > 1$ changed slightly with thermal stability, except at $X/H = 1.5$.

Vertical distributions of concentration fluctuation are shown in Fig. 16. The effect of thermal stratification is clear at $Z/H = 0.5$ and $X/H = 0.5, 1.0$ and 1.5 . Concentration fluctuations show a pattern similar to the mean concentration. The effect of stability on concentration fluctuations was slight at $Z/H < 0.75$, near the stack, but increased rapidly through $Z/H = 1.0$ and generally decreased at $Z/H > 1.0$. Fluctuations in its lower portion were close to zero, because the plume spread was so narrow that only very small eddies could be generated within it. Concentration fluctuations were less than 1 at $Z/H < 1$, but reached a maximum of $I = 2$ near the roof surface at $X/H = 0.25$. The fluctuations slowly decreased at distances far downwind of the stack.

The features pollutant concentration distributions under the three different thermal stabilities are as follows:

(a) Stable atmospheric stability:

The vertical spread of plume under stable conditions observed wider than under neutral and unstable conditions. The higher maximum concentrations observed under stable conditions. The Vertical concentration distributions increased under stable conditions in the near wake of the building. This is because of stable stability suppresses vertical motion and reduces vertical spreading. This result shows that air changes in the near wake were affected mainly by turbulent mixing between the wake and free stream regions. The concentration levels close to the ground surface were higher than under unstable and neutral conditions. Lateral concentration distributions was lower than under unstable and neutral conditions at $X/H = 0.25$ and 0.5 in the near wake of the building. This is because of the lateral flow carries the contaminant away from the building. Those results are similar to what observed by Santos et al. (2009). Concentration fluctuations under stable conditions at $Z/H > 1$ were larger than those under neutral and unstable conditions. This is because of the strong influence of turbulent energy in the region.

Fig. 14. Vertical profiles of mean concentration, K .

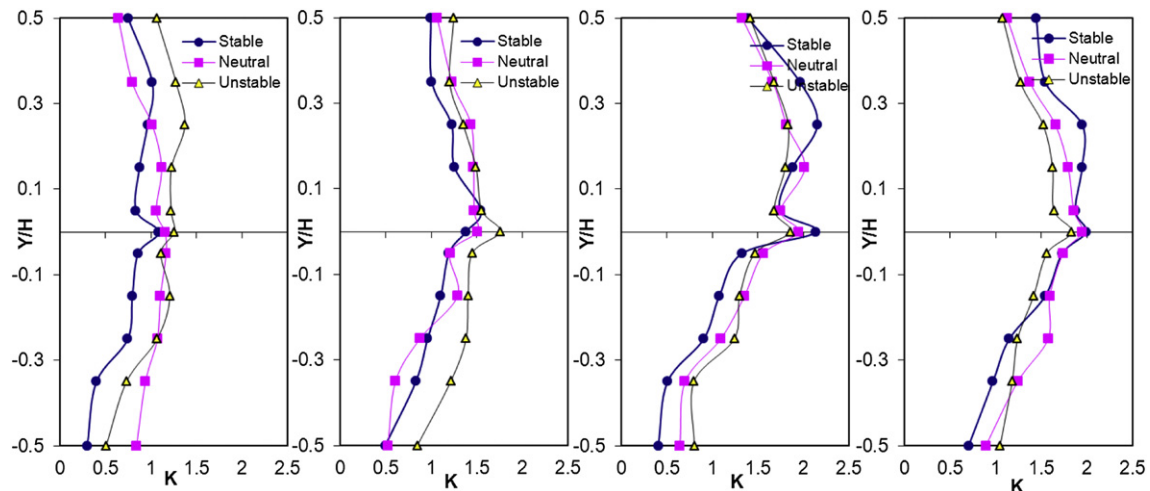


Fig. 15. Lateral profiles of mean concentration, K at $Z/H = 0.5$.

(b) Unstable atmospheric stability:

The vertical spread of plume under unstable conditions observed narrower than under neutral conditions. The maximum concentrations observed under unstable conditions. Concentration distributions decreased under unstable conditions in the near wake region. Due to the larger longitudinal turbulent intensity observed under unstable conditions. The lateral spreading of the plume was larger. This is because of the higher longitudinal turbulent yielding larger turbulent spread of the plume, the size of the region of flow distributed by building also affects the plume spreading. The concentration levels close to the ground surface were lower than under stable and neutral conditions. Pollutant concentration in the near wake at $Z/H < 0.75$ under unstable conditions was lower than that under neutral and stable conditions. This is because vertical velocity was lower in this region. However, concentration over the building roof ($Z/H > 1$) at $X/H = 1.5$ (far from the stack) under unstable conditions was higher than that under neutral and stable conditions, consistent with Ogawa et al. (1974). This is because the turbulent diffusion of concentration is reduced in this region.

Smaller concentration fluctuations were observed under unstable conditions in the upper portion of the near wake.

(c) Neutral atmospheric stability:

The vertical spread of plume under unstable conditions observed wider than under unstable conditions. Concentration distributions were higher than under unstable conditions and lower than under stable conditions in the near wake region. The concentration levels were larger closer to the ground surface than under unstable conditions. This is because of the concentration levels are quite uniform. Lateral concentration distributions was higher at $X/H = 0.25$ and 0.5 and lower at $X/H = 1.0$ and 1.5 than under stable conditions in the near wake of the building. This is because of the effects of lateral flow distributions in the in the near wake of the building. Concentration fluctuations under neutral conditions close to the ground surface were higher than those under unstable conditions at $X/H = 0.25$ and 0.5 .

This result shows that the effect of thermal stability in the near wake of the building is to increase concentration under stable

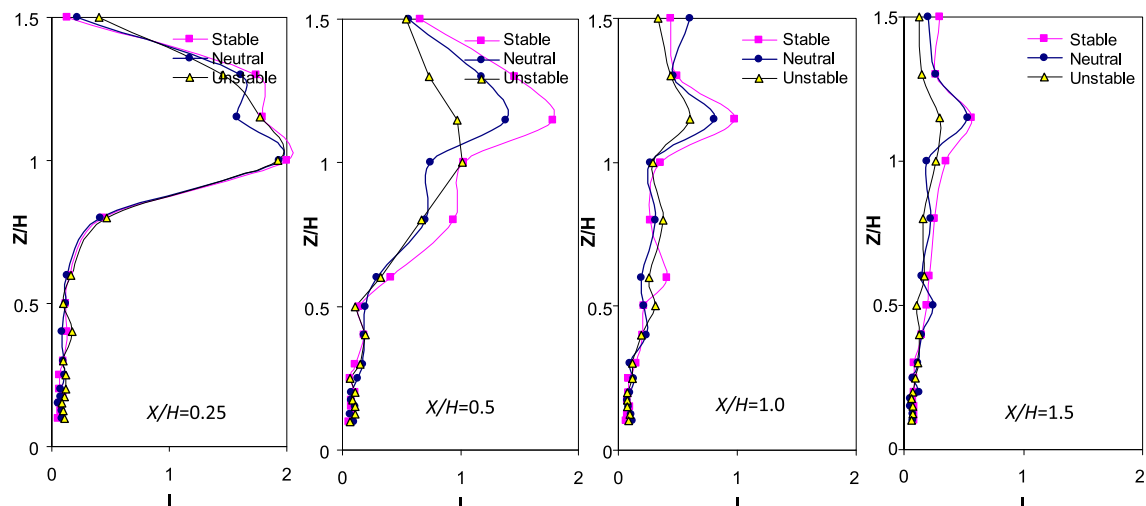


Fig. 16. Vertical profiles of concentration fluctuation intensity, I .

conditions, and decrease it under unstable conditions. An increase in longitudinal turbulence intensity decreases concentration with downwind distance under unstable conditions. An increase in vertical turbulence intensity under stable conditions increases the concentration. Variations in stability conditions produce variations in the conditions of the approach concentration fluctuations in the near wake region.

5. Conclusions

A wind tunnel study was conducted to investigate the effect of thermal stability on flow and dispersion of emissions from a rooftop stack in the near wake region. The experiments were carried out for varying stratifications using stack located of the building roof. The main conclusions of the study are:

(1) For mean velocity measurement:

When thermal stability was strong, wind velocity in the lower part of the near wake of the building was very low. Flow patterns appeared symmetric in both edges of the lee side of the building at $Z/H = 0.5$. When atmospheric stratification is stable, vertical velocity increases in the near wake of the building. The stable profiles show decreased longitudinal velocity at the lower height near the ground surface. When this stratification is unstable, vertical velocity decreases and longitudinal velocity increases in the near wake. The flow separation in the wake region is smaller than under stable and neutral conditions. The length of the recirculation region in the wake region is longer. The lateral profiles of longitudinal velocity decreased at $Z/H = 0.5$. When atmospheric stratification, the flow separation in the wake region is higher than under stable condition. Vertical velocity was lower than under stable and unstable conditions.

(2) For turbulent measurements

The effect of thermal stability on longitudinal and vertical turbulence velocity was much larger with increasing distance from the building model. Peaks of longitudinal and vertical turbulence intensities were observed in its upper portion, near the stack. When atmospheric stratification is stable, vertical turbulence velocity increases in the near wake of the building. When this stratification is unstable, longitudinal turbulence velocity increases in the near wake of the building. The longitudinal and vertical turbulence intensities observed to be lower than those in stable and unstable conditions in the near wake of the building.

(3) Pollutant concentration measurements

The vertical distributions of concentration clearly show downwash in the near wake of the building. Peak concentrations of thermal stability were observed at $Z/H = 1$. The maximum concentration was near the stack emission on the roof surface. The minimum concentration on the roof surface was far from the stack emission. The effect of thermal stability in the near wake of the building is to increase concentration under stable conditions, and decrease it under unstable conditions. The vertical spread of plume under stable conditions observed wider than under neutral and unstable conditions. The lateral spreading of the plume was larger. The higher maximum concentrations observed under stable conditions. Concentration fluctuations under stable conditions at $Z/H > 1$ were larger than those under unstable conditions. Smaller concentration fluctuations were observed under unstable conditions in the upper portion of the near wake.

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References

- Ahmadi, G., Li, A., 2000. Computer simulation of particle transport and deposition near a small isolated building. *Journal of Wind Engineering and Industrial Aerodynamics* 84, 23–46.
- Arya, S.P., 1999. *Air Pollution Meteorology and Dispersion*. Oxford University Press, Inc, Oxford.
- ASHRAE, 2007. Building air intake and exhaust design (Chapter 44). In: *ASHRAE Handbook – Heating, Ventilating, and Air-conditioning Applications*. American Society of Heating, Refrigerating and Air-conditioning Engineers, Atlanta, USA.
- ASHRAE, 2009. Airflow around buildings (Chapter 24). In: *ASHRAE Handbook – Fundamentals*. American Society of Heating, Refrigerating and Air-conditioning Engineers, Atlanta, USA.
- ASHRAE, 2011. Building Air Intake and Exhaust Design (Chapter 45). In: *ASHRAE Applications Handbook*. American Society of Heating, Refrigerating and Air-conditioning Engineering Inc, Atlanta, USA.
- Castro, I.P., Robins, A.G., 1977. The flow around a surface mounted cube in uniform and turbulent streams. *Journal of Fluid Mechanics* 79, 307–335.
- Chavez, M., Hajra, B., Stathopoulos, T., Bahloul, A., 2011. Near-field pollutant dispersion in the built environment by CFD and wind tunnel simulations. *Journal of Wind Engineering and Industrial Aerodynamics* 99 (4), 30–39.
- Flowe, A.C., Kumar, A., 2000. Analysis of velocity fields and dispersive cavity parameters as a function of building width to building height ratio using a 3-D computer model for squat buildings. *Journal of Wind Engineering and Industrial Aerodynamics* 86, 87–122.
- Gupta, A., Stathopoulos, T., Saathoff, P., 2012. Wind tunnel investigation of the downwash effect of a rooftop structure on plume dispersion. *Atmospheric Environment* 46, 496–507.
- Hajra, Stathopoulos, T., 2012. A wind tunnel study of the effect of downstream buildings on near-field pollutant dispersion. *Building and Environment* 52, 19–31.
- Hajra, Stathopoulos, T., Bahloul, A., 2011. The effect of upstream buildings on near-field pollutant dispersion in the built environment. *Atmospheric Environment* 45, 4930–4940.
- Higson, H.L., Griffiths, R.F., Jones, C.D., Biltoft, C., 1995. Effect of atmospheric stability on concentration fluctuations and wake retention times for dispersion in the vicinity of an isolated building. *Environmetrics* 6, 571–581.
- Hosker Jr., R.P., 1984. Flow and diffusion near obstacles. In: *Randerson, D. (Ed.), Atmospheric Science and Power Production*. United States Department of Energy, pp. 241–326.
- Hoult, D.P., Weil, J.C., 1972. Turbulent plume in a laminar cross flow. *Atmospheric Environment* 6 (8), 513–530.
- Kothari, K.M., Peterka, J.A., Meroney, R.N., 1985. Wind flow patterns about buildings. *Journal of Wind Engineering and Industrial Aerodynamics* 21, 21–38.
- Lape, J.F., 1987. *A Wind Tunnel Study of Relaxation Schemes Used for Buoyant Plume-rise Modeling*. North Carolina St. Univ. M.Sc.
- Lateb, M., Masson, C., Stathopoulos, T., Bédard, C., 2011. Effect of stack height and exhaust velocity on pollutant dispersion in the wake of a building. *Atmospheric Environment* 45, 5150–5163.
- Li, W.W., Meroney, R.N., 1983. Gas dispersion near a cubical model building. Part I mean concentration measurement. *Journal of Wind Engineering and Industrial Aerodynamics* 77 & 78, 619–630.
- Lowery, Kirk P., Jacko, Robert B., 1996. A wind tunnel study into the effects of raised intakes and parapets on fresh air intake contamination by a rooftop stack. *Journal of the Air & Waste Management Association* 46 (9), 847–852.
- Mavroidis, I., Griffiths, R.F., Jones, C.D., Biltoft, C., 1999. Experimental investigation of the residence of contaminants in the wake of an obstacle under different stability conditions. *Atmospheric Environment* 33, 939–949.
- Ogawa, Y., Griffiths, R., Hoydush, W.G., 1974. A wind tunnel study of sea breeze effects. *Boundary Layer Meteorology* 8, 141–161.
- Ogawa, Y., Diosey, P.G., Uehara, K., Ueda, H., 1981. A wind tunnel for studying the effects of thermal stratification in the atmosphere. *Atmospheric Environment* 15 (5), 807–821.
- Olesen, H.R., Berkowicz, R., Ketzel, M., Løfstrøm, P., 2009. Validation of OML, AERMOD/PRIME and MISCAM using the Thompson wind-tunnel dataset for simple stack-building configurations. *Boundary Layer Meteorology* 2009 (131), 73–83.
- Panofsky, H.A., Dutton, J.A., 1984. *Atmospheric Turbulence*. John Wiley & Sons, New York.
- Petersen, R.L., Cochran, B.C., Carter, J.J., 2002. Specifying exhaust and intake systems. *ASHRAE Journal* 4, 30–35.
- Saathoff, P., Gupta, A., Stathopoulos, T., 2009. Contamination of fresh air intakes due to downwash from a rooftop structure. *Journal of the Air & Waste Management Association* 59, 343–353.

- Saathoff, P.J., Stathopoulos, T., Dobrescu, M., 1995. Effects of model scale in estimating pollutant dispersion near buildings. *Journal of Wind Engineering and Industrial Aerodynamics* 54–55 (2), 549–559.
- Santos, J.M., Griffiths, R.F., Reis Jr, N.C., Mavroidis, I., 2009a. Experimental investigation of averaging time effects on building influenced atmospheric dispersion under different meteorological stability conditions. *Building and Environment* 44 (6), 1295–1305.
- Santos, J.M., Reis, N.C., Griffiths, R.F., Mavroidis, I., 2009b. Numerical simulation of flow and dispersion around an isolated cubical building: The effect of the atmospheric stratification. *Atmospheric Environment* 43 (34), 5484–5492.
- Schulman, L., Scire, J., 1991. The effect of stack height, exhaust speed, and wind direction on concentration from rooftop stack. *ASHRAE Transactions* 97, 573–582.
- Smith, W.S., Reisner, J.M., Kao, C.-Y.J., 2001. Simulations of flow around a cubical building: comparison with towing-tank data and assessment of radioactively induced thermal effects. *Atmospheric Environment* 35, 3811–3821.
- Snyder, W.H., 1981. Guideline for Fluid Modeling of Atmospheric Diffusion. EPA–600/8-81-009.
- Snyder, W.H., 1985. Fluid modeling of pollutant transport and diffusion in stably stratified flows over complex terrain. *Annual Review of Fluid Mechanics* 17, 239–266.
- Snyder, W.H., 1994. Some observations of the influence of stratification on diffusion in building wake. In: Castro, I.P., Rockliff, N.J. (Eds.), *Stably Stratified Flows and Dispersion over Topography*. Clarendon Press, Oxford, England, pp. 301–324.
- Snyder, W.H., 2005. Streamline patterns around buildings deduced from wind tunnel measurement. In: *Proceedings of PHYSMOD*. London, ON. August 24e26.
- Snyder, W.H., Lawson, R.E., 1991. Fluid modeling simulation of stack-tip downwash for neutrally buoyant plumes. *Atmospheric Environment* 25A (12), 2837–2850.
- Tennekes, H., Lumley, J.L., 1994. *A First Course in Turbulence*. The MIT Press.
- Thompson, R.S., Lombardi, D.J., 1977. Dispersion of Rooftop Emission from Isolated Building: a Wind Tunnel Study. EPA-600/4-77-006, United States Environmental Protection Agency, 36 pp.
- Thompson, R., 1993. Building amplification factors for sources near buildings: a wind tunnel study. *Atmospheric Environment* 27, 2313–2325.
- Uehara, K., Murakami, S., Oikawa, S., Wakamatsu, S., 2000. Wind tunnel experiments on how thermal stratification affects flow in and above urban street canyons. *Atmospheric Environment* 34 (10), 1553–1562.
- White, B.R., Stein, W., 1990. Wind-tunnel studies of variable stack heights for a low profile building. *Journal of Wind Engineering and Industrial Aerodynamics* 36 (1–3), 675–687.
- Wilson, D.J., 1979. Flow patterns over flat-roofed buildings and application to exhaust stack design. *ASHRAE Transactions* 85, 284–295.
- Wilson, D.J., Fabris, I., Chen, J., Ackerman, M., 1998. Adjacent building effects on laboratory fume hood exhaust stack design. *ASHRAE Transactions* 104, 1012–1027.
- Yang, B.T., Meroney, R.N., 1970. Gaseous Dispersion into Stratified Building Wakes. AEC Rept. No. C00-2053-3.
- Yassin, M.F., Kato, S., Ooka, R., Takahashi, T., Kouno, R., 2005. Field and wind-tunnel study of pollutant dispersion in a built-up area under various meteorological conditions. *Journal of Wind Engineering and Industrial Aerodynamics* 93, 361–382.
- Zegadi, R., Ayrault, M., Mejean, P., 1994. Effect of a two-dimensional low hill in thermally neutral and stable stratified turbulent boundary layer. *Atmospheric Environment* 28 (1), 1871–1879.
- Zhang, Y.Q., Arya, S.P., Snyder, W.H., 1996. A comparison of numerical and modeling of stable atmospheric flow and dispersion around a cubical building. *Atmospheric Environment* 30, 1327–1345.